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METHOD OF OBTAINING A GRID OF BLADES OF AN AIRCRAFT THRUST REVERSING DEVICE

Abstract

A method of obtaining a grid of blades of an aircraft thrust reversing device by producing U-shape elements each including an integral central part corresponding to a blade and first and second branches each corresponding to at least a part of a longitudinal wall of the grid of blades and assembling which includes connecting the first branches of the various U-shape elements to one another in such a manner as to form a first longitudinal wall and connecting the second branches of the various U-shape elements to one another in such a manner as to form a second longitudinal wall.

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Background/Summary

CROSS-REFERENCES TO RELATED APPLICATIONS

[0001] This application claims the benefit of French Patent Application Number FR2402299 filed on Mar. 7, 2024, the entire disclosure of which is incorporated herein by way of reference.

FIELD OF THE INVENTION

[0002] The present invention relates to a method of obtaining a grid of blades of an aircraft thrust reversing device.

BACKGROUND OF THE INVENTION

[0003] In an embodiment that can be seen in FIGS. **1** to **3**, an aircraft **10** comprises a plurality of propulsion assemblies **12** positioned under the wings **14** of the aircraft **10** and connected to the latter by pylons **16**. Each propulsion assembly **12** comprises an engine **18** positioned inside a nacelle **20**. The engine **18** comprise a fan that has a rotation axis **A18**.

[0004] In the remainder of the description a longitudinal direction is parallel to the rotation axis **A18**. A longitudinal plane is a plane containing the rotation axis **A18** and a transverse plane is a plane perpendicular to the rotation axis **A18**. The front/upstream and rear/downstream concepts refer to the direction of flow of a flow of air in the nacelle **20**, flowing from the front (upstream) to the rear (downstream).

[0005] The nacelle **20** has an approximately tubular shape and with the engine **18** delimits an annular duct **22** in which flows a secondary flow of air. The nacelle **20** comprises an upstream section **24**, called the air intake, a median section **26** in which the fan of the engine **18** is positioned, and a downstream section **28** that has a trailing edge **30**.

[0006] The nacelle **20** comprises a thrust reversing device **34** positioned at the level of the downstream section **28** and configured to occupy an activated state in which it diverts at least some of the secondary flow of air circulating in the annular duct **22** outward and upstream of the nacelle **20** and an inactivated state in which it does not divert the secondary flow of air circulating in the annular duct **22**.

[0007] The thrust reversing device **34** comprises at least one mobile part **36** that enables at least one lateral opening **38** (visible in FIG. **2**) through the nacelle **20** to be formed.

[0008] The thrust reversing device **34** comprises a diversion system **40** configured to occupy a retracted position when the thrust reversing device **34** is in the inactivated state, in which the diversion system **40** does not interfere with the secondary flow of air, and a deployed position when the thrust reversing device **34** is in the activated state, in which the diversion system **40** interferes with the secondary flow of air and orients it toward the lateral opening **38**.

[0009] The thrust reversing device **34** also comprises an orientation system **42** configured to orient the flow of air diverted by the diversion system **40**. In accordance with one embodiment, the orientation system **42** comprises a plurality of grids of blades **44** called cascades positioned at the level of each lateral opening **38**.

[0010] In accordance with an embodiment that can be seen in FIGS. **4** and **5** each grid of blades **44** includes longitudinal walls **46** positioned in approximately longitudinal planes and blades **48** positioned in approximately transverse planes. The longitudinal walls **46** and the blades **48** delimit cells **50**.

[0011] The blades **48** have a profile for diverting upstream the flow of air exiting via each lateral opening **38**. To this end, each blade **48** is curved and has a depth **P**, as depicted in FIG. **5**.

[0012] In accordance with one arrangement the grid of blades **44** comprises five longitudinal walls **46** delimiting four rows of blades **48**, sixteen blades **48** in each row and sixty-four cells **50**.
[0013] Given their geometry and notably the small depth P of the blades **48**, the grids of blades **44** include a large number of cells **50** and therefore have a relatively high mass, which for an aircraft represents a penalty in terms of onboard mass and energy consumption.
[0014] As depicted in FIGS. **6** and **7**, a grid of blades **44** is generally made by molding a composite material, using for each cell **50** a core **52** configured to expand.
[0015] In accordance with one operating mode a method of obtaining a grid of blades **44** comprises a step of stacking layers of fibers to obtain first preforms **46'**, one for each longitudinal wall **46**, and second preforms **48'**, one for each blade **48**. As depicted in FIG. **6**, the first preforms **46'** are substantially planar. The second preforms **48'** have an H-shape section provided at each end with first and second flanges **48.1'**, **48.2'** that are intended to be pressed against first preforms **46'**.
[0016] The method of obtaining a grid of blades **44** comprises, for each row of blades, a step of placing alternately second preforms **48'** and cores **52** between two substantially parallel (i.e., +/-10%) first preforms **46'**, the first flanges **48.1'** of all the second preforms **48'** being pressed against the first preform **46'** situated on the left, the second flanges **48.2'** of all the second preforms **48'** being pressed against the first preform **46'** situated on the right. When all the first and second preforms **46'**, **48'** have been positioned, the method comprises a consolidation or polymerization step to connect the first and second preforms **46'**, **48'** together during which the assembly is compressed and subjected to an increase in temperature.

[0017] The rise in temperature causes expansion of the cores **52** which compress the second preforms **48'**. After the consolidation or polymerization step the grid of blades **44** obtained and the cores **52** are cooled, which causes the cores **52** to shrink. The method then comprises a step of extraction from the mold during which the cores **52** are extracted from the grid of blades **44**.
[0018] Given the relatively small depth P of the blades **48**, the cores **52** compress them correctly during the consolidation or polymerization step, which enables control of the structural integrity of the grid of blades **44** obtained.

[0019] This method enables a good compromise to be achieved between providing pressure by expansion and preserving a capacity for extraction from the mold when the blades have a small depth. This method, suitable for blades with a small depth, is not suitable if the blades have a large depth, the cores not expanding sufficiently to compress the preforms. To enable extraction from the mold of a grid of blades that includes blades with a large depth, it is necessary to provide relatively complex sliding cores that are difficult or even impossible to extract.

[0020] More globally, because of their geometry, grids of blades are complex to produce.

SUMMARY OF THE INVENTION

[0021] The present invention aims to remedy some or all of the disadvantages of the prior art.
[0022] To this end, the invention has for object a method of obtaining a grid of blades of an aircraft thrust reversing device, said grid of blades including at least first and second longitudinal walls oriented in a longitudinal direction and blades positioned between said first and second longitudinal walls, connected to the latter and oriented in a transverse direction.
[0023] According to the invention the method comprises: [0024] a step of production of U-shape elements, one for each blade, each U-shape element including a central part corresponding to one of the blades, a first branch corresponding to at least a part of the first longitudinal wall and a second branch corresponding to at least a part of the second longitudinal wall, the central part and the first and second branches forming only one and the same part, [0025] a step of assembling the U-shape preforms consisting in connecting the first branches of the various U-shape preforms to one another in such a manner as to form the first longitudinal wall and connecting the second branches of the various U-shape preforms to one another in such a manner as to form the second longitudinal wall.

[0026] Producing a grid of blades by assembling U-shape elements enables blades with large

depths to be obtained whilst preserving simple tooling because of the geometry with no undercut of the U-shape elements.

[0027] In accordance with another feature, at least one U-shape preform has an interior face with no undercut. Additionally, the step of producing the U-shape preforms comprises, before the compression phase and for at least one U-shape preform, a draping phase during which layers of fibers are positioned on a first mold shaped like the interior face of the U-shape preform.

[0028] In accordance with another feature, each U-shape element is a U-shape preform made of a composite material, the production step including for at least one U-shape preform at least one compression phase and at least one curing or polymerization phase during which said U-shape preform is compressed and at least partially cured or polymerized.

[0029] In accordance with another feature, each U-shape preform is partially cured or polymerized during the production step, which comprises a phase of partial curing or polymerization, a phase of final curing or polymerization being carried out subsequently and at the latest during the assembly step.

[0030] In accordance with another feature, each U-shape preform is totally cured or polymerized during the production step.

[0031] In accordance with another feature, during the production step the U-shape preform is compressed between on the one hand the first mold and on the other hand at least one matching mold and/or at least one compression bladder.

[0032] In accordance with another feature, during the draping phase, the layers of fibers are positioned in such a manner as to have fibers mostly oriented in the transverse direction at the level of the central part and fibers mostly oriented in the longitudinal direction at the level of each of the first and second branches.

[0033] In accordance with another feature, during the draping phase at least one layer of fibers is positioned in such a manner as to extend over the central part and at least over one of the first and second branches.

[0034] In accordance with another feature, for each U-shape element the first and second branches are spaced by a distance apart that is substantially constant over their length. Additionally, the distances apart separating the first and second branches of the various U-shape elements decrease from one end to the other of the grid of blades so as to be able to form the first and second longitudinal walls by respectively stacking the first and second branches of the various U-shape elements.

[0035] In accordance with another feature, the first and second branches of the various U-shape elements have free ends situated at the level of one of the ends of the grid of blades.

[0036] In accordance with another feature, for at least one of the first and second U-shape elements each of the first and second branches comprises a first section that extends from the central part, a second section that extends from the first section as far as a free end of the first or second branch and an inclined section connecting the first and second sections. Additionally, the first sections of the first and second branches of the first and second U-shape elements are spaced by a first distance, the second sections of the first and second branches being spaced by a second distance greater than the first distance, the first sections of the first and second branches of the second U-shape element being positioned between the second sections of the first and second branches of the first U-shape element and connected to said second sections.

[0037] In accordance with another feature, for at least one U-shape element the central part comprises a first part forming a body of a blade and a second part substantially perpendicular to the first part forming a flange of the blade. In accordance with another feature, the second part extends over all the central part and at least partially along at least one of the first and second branches.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0038] Other features and advantages emerge from the following description of the invention given by way of example only with reference to the appended drawings, in which:

[0039] FIG. **1** is a perspective view depicting one embodiment of an aircraft and a propulsion assembly,

[0040] FIG. **2** is a lateral view depicting a prior art embodiment of a propulsion assembly,

[0041] FIG. **3** is a schematic section depicting a prior art embodiment of a propulsion assembly equipped with a thrust reversing device in the inactivated step,

[0042] FIG. **4** is a perspective view depicting a prior art embodiment of a grid of blades,

[0043] FIG. **5** is a longitudinal section on the plane P-V of the grid of blades that can be seen in FIG. **4**,

[0044] FIG. **6** is a schematic representation depicting a prior art embodiment of a step of placing longitudinal walls, blades and cores,

[0045] FIG. **7** is a view from above depicting a prior art embodiment of a grid of blades during a consolidation or polymerization step,

[0046] FIG. **8** is a perspective view depicting one embodiment of a grid of blades of the invention,

[0047] FIG. **9** is a longitudinal section on the plane P-IX of the grid of blades that can be seen in FIG. **8**,

[0048] FIG. **10** is a perspective view depicting one embodiment of a blade of the invention,

[0049] FIG. **11** is a perspective view depicting one embodiment of a part of a grid of blades of the invention during assembly,

[0050] FIG. **12** is a cross section depicting one embodiment of a blade profiled member of the invention positioned on a mold,

[0051] FIG. **13** is a longitudinal section depicting one embodiment of a blade profiled member of the invention positioned between a mold and a matching mold,

[0052] FIG. **14** is a view from above depicting one embodiment of end plates of the invention,

[0053] FIG. **15** is a view from above depicting one embodiment of a part of a grid of blades of the invention during assembly,

[0054] FIG. **16** is a view from above depicting one embodiment of an assembled grid of blades of the invention,

[0055] FIG. **17** is a view from above depicting another embodiment of a part of a grid of blades of the invention during assembly.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

[0056] In accordance with one application, an aircraft comprises at least one thrust reversing device including at least one grid of blades **60**.

[0057] In accordance with an embodiment that can be seen in FIGS. **8** to **10**, a grid of blades **60** includes at least first and second longitudinal walls **62**, **62'** positioned in approximately longitudinal planes and blades **64** positioned between said first and second longitudinal walls **62**, **62'**, connected to the latter and positioned in approximately transverse planes, the longitudinal walls **62** and the blades **64** delimiting cells **66**.

[0058] Each longitudinal wall **62** extends between inner and outer edges **62.1**, **62.2**. Each blade **64** extends between inner and outer edges **64.1**, **64.2**. In accordance with one arrangement, the inner edges **62.1**, **64.1** of the longitudinal walls **62** and the blades **64** are positioned at the level of a first surface forming an interior surface F**60** of the grid of blades **60**, the exterior edges **62.2**, **64.2** of the longitudinal walls **62** and the blades **64** being positioned at the level of a second surface forming an exterior surface F**60'** of the grid of blades **60**. In accordance with one configuration, the inner and outer surfaces F**60**, F**60'** are curved to adapt to the curvature of the inner cell. Of course, the invention is not limited to this geometry of the outer and inner surfaces F**60**, F**60'**.

[0059] The blades **64** have inner and outer edges **64.1**, **64.2** substantially parallel to one another

and oriented in a transverse direction substantially perpendicular to the longitudinal direction.

[0060] In accordance with one configuration, each blade **64** has a substantially constant section in the transverse direction. The blades **64** can have identical sections. Alternatively, as depicted in FIG. **9**, the blades **64** can have different sections from one blade to another.

[0061] In accordance with an embodiment that can be seen in FIG. **10**, each blade **64** comprises a body **68** that extends between a first edge **68.1** corresponding to the inner edge **64.1** of the blade **64** and a second edge **68.2** and a flange **70** situated in line with the body **68** that extends between a first edge **70.1** corresponding to the outer edge **64.2** of the blade **64** and a second edge **70.2** rigidly attached to the second edge **68.2** of the body **68**. In a longitudinal plane the flange **70** is substantially perpendicular to the body **68**. For each blade **64** the body **68** and the flange **70** form one and the same part.

[0062] In accordance with one configuration, the flange **70** is substantially plane and situated at the level of the outer surface **F60'** of the grid of blades **60**.

[0063] The blade **64** has a depth **P** that corresponds to a maximum distance separating a plane **PL** passing through the inner and outer edges **64.1**, **64.2** of the blade **64** and a generatrix **C** of the body **68** farthest from the plane **PL**.

[0064] In accordance with one configuration, the depth **P** of the blade **64** is relatively large. By relatively large is meant that the depth **P** does not allow extraction from the mold of the grid of blades using simple cores in accordance with a prior art method that can be seen in FIGS. **6** and **7**.

[0065] In accordance with one arrangement, the grid of blades **60** comprises three longitudinal walls **62** defining two rows of blades **72**, **72'** and four blades **64** per row of blades, the longitudinal walls **62** and the blades delimiting eight cells **66**.

[0066] Given the depth **P** of the blades **64**, it is possible to reduce their number, which tends to reduce the mass of the grid of blades **60**.

[0067] Generally speaking, a grid of blades **60** comprises a plurality of rows of blades **72**, **72'**. However, a grid of blades **60** could comprise only one row of blades **72**.

[0068] Whatever the embodiment, the grid of blades **60** comprises edge longitudinal walls corresponding to the farthest apart longitudinal walls **62**. It extends between a front end **60.1** situated approximately in a first transverse plane and a rear end **60.2** situated approximately in a second transverse plane.

[0069] In accordance with one embodiment, the grid of blades **60** comprises at least one end plate **74**, situated at the level of the front or rear end **60.1**, **60.2**, connecting the edge longitudinal walls and positioned at the level of the outer surface **F60'** of the grid of blades **60**. In accordance with a configuration that can be seen FIGS. **14** and **15**, each end plate **74** is a substantially rectangular strip of material that has long sides substantially parallel to the transverse direction. At least one end plate **74** comprises at least one through-orifice **74.1** configured for fixing the grid of blades **60** to a structure of an aircraft.

[0070] In accordance with one arrangement, the grid of blades **60** comprises two end plates **74**, **74'** positioned at the front and rear ends **60.1**, **60.2** of the grid of blades **60**.

[0071] In accordance with an operating mode that can be seen in FIGS. **11** to **17**, a method of obtaining a grid of blades comprises: [0072] a step of producing U-shape preforms **76.1** to **76.4** made of a composite material, one for each blade **64**, each U-shape preform including a central part **78** corresponding to one of the blades **64**, a first branch **80** corresponding to at least a part of a first longitudinal wall **62**, and a second branch **82** corresponding to at least a part of a second longitudinal wall **62'**, the central part **78** and the first and second branches **80**, **82** forming only one and the same part, [0073] a step of assembling the U-shape preforms **76.1** to **76.4** in such a manner as to obtain a row of blades by connecting the first branches **80** of the various U-shape preforms **76.1** to **76.4** to one another in order to form the first longitudinal wall **62** and connecting the second branches **82** of the various U-shape preforms **76.1** to **76.4** to one another in order to form the second longitudinal wall **62'**.

[0074] When the grid of blades **60** comprises a plurality of rows of blades **72, 72'**, the method of obtaining a grid of blades comprises a step of assembling the various rows of blades **72, 72'** in such a manner as to obtain the grid of blades **60**.

[0075] When the grid of blades **60** comprises at least one end plate **74, 74'**, the method of obtaining a grid of blades comprises a step of producing each end plate **74, 74'** and a step of assembling each end plate **74, 74'** consisting in connecting it to at least one U-shape preform **76.1 to 76.4**.

[0076] Each U-shape preform **76.1 to 76.4** made of a composite material comprises fibers buried in a thermoset or thermoplastic resin matrix. Each end plate **74, 74'** is made of a composite material comprising fibers buried in a thermoset or thermoplastic resin matrix. In accordance with one configuration, the U-shape preforms **76.1 to 76.4** and the end plates **74, 74'** are made of the same composite material.

[0077] The steps of assembling the U-shape preforms **76.1 to 76.4**, the rows **72, 72'** and the end plates **74, 74'** can be distinct steps. Alternatively, the steps of assembling the

[0078] U-shape preforms **76.1 to 76.4**, the rows **72, 72'** and the end plates **74, 74'** can be carried out at the same time and constitute only one and the same step.

[0079] When the U-shape preforms **76.1 to 76.4** and the end plates **74, 74'** if any each comprise a thermoset resin matrix, each assembly step consists in connecting the U-shape preforms **76.1 to 76.4** and the end plates **74, 74'** if any with fixing elements such as rivets or screws for example, possibly using fishplates. Alternatively, the U-shape preforms **76.1 to 76.4** and the end plates **74, 74'** if any are connected by gluing or co-curing.

[0080] When the U-shape preforms **76.1 to 76.4** and the end plates **74, 74'** if any each comprise a thermoplastic resin matrix each assembly step consists in connecting the U-shape preforms **76.1 to 76.4** and the end plates **74, 74'** if any with fixing elements, such as rivets or screws for example, possibly using fishplates. Alternatively, the U-shape preforms **76.1 to 76.4** and the end plates **74, 74'** if any are connected by gluing, co-curing, co-consolidation or welding.

[0081] In accordance with a first operating mode, each U-shape preform **76.1 to 76.4** is partly cured or polymerized during the production step, which comprises a phase of partly polymerizing or curing, a phase of final polymerization or curing being carried out subsequently and at the latest during the assembly step.

[0082] In accordance with a second operating mode, each U-shape preform **76.1 to 76.4** is totally cured or polymerized during the production step that comprises a phase of complete polymerization or curing of the U-shape preform **76.1 to 76.4**.

[0083] In accordance with a first embodiment that can be seen in FIGS. **15, 16**, each of the first and second branches **80, 82** of each U-shape preform **76.1 to 76.4** is plane or curved. For each U-shape preform **76.1 to 76.4**, the first and second branches **80, 82** are spaced by a substantially constant distance apart over their length. In accordance with this first embodiment, the distances apart separating the first and second branches **80, 82** of the various U-shape preforms **76.1 to 76.4** decrease from one end to the other of the grid of blades **60** so as to be able to form the first and second longitudinal walls **62, 62'** by respectively stacking the first and second branches **80, 82** of the various U-shape preforms **76.1 to 76.4**. In accordance with one arrangement, the distances apart separating the first and second branches **80, 82** of the U-shape preforms **76.1 to 76.4** decrease from the rear end **60.2** to the front end **60.1** of the grid of blades **60**. In accordance with this first embodiment, the distances apart separating the first and second branches **80, 82** of the various U-shape preforms **76.1 to 76.4** are determined so that for the first longitudinal wall **62** the first branches **80** are stacked against one another and connected to one another and so that for the second longitudinal wall **62'** the second branches **82** are stacked against one another and connected to one another.

[0084] In accordance with this first embodiment, the first and second branches **80, 82** of the various U-shape preforms **76.1 to 76.4** have two free ends **80.1, 82.1** situated at the level of one of the ends of the grid of blades **60**, notably the front end **60.1** of the grid of blades **60**. In accordance with this

first embodiment, the grid of blades **60** comprises first, second, third and fourth U-shape preforms **76.1** to **76.4** offset in the longitudinal direction. The first preform **76.1** has a central part **78** situated at the level of the rear end **60.2** of the grid of blades **60** and first and second branches **80**, **82** that extend from the rear end **60.2** to the front end **60.1** of the grid of blades **60**. The first and second branches **80**, **82** of the second, third and fourth U-shape preforms have free ends **80.1**, **82.1** situated like those of the first and second branches **80**, **82** of the first U-shape preform **76.1** at the level of the front end **60.1** of the grid of blades **60**. Additionally, the end plate **74** situated at the level of the front end **60.1** has ends connected at least to the first and second branches **80**, **82** of the fourth U-shape preform **76.4** having the central part **78** nearest the front end **60.1** of the grid of blades **60**.

[0085] In a second embodiment that can be seen in FIGS. **11** and **17**, for at least one of the first and second U-shape preforms **76.1** to **76.4** their first and second branches **80**, **82** comprise a first section **T1** that extends from the central part **78**, a second section **T2** that extends from the first section **T1** to a free end **80.1**, **82.1** of the first or second branch **80**, **82**, and an inclined section **84** connecting the first and second sections **T1**, **T2**.

[0086] In accordance with the second embodiment, the first sections **T1** of the first and second branches **80**, **82** are spaced by a first distance **D1** and the second sections **T2** of the first and second branches **80**, **82** of the first and second U-shape preforms **76.1** to **76.4** are spaced by a second distance **D2** greater than the first distance **D1**. All the U-shape preforms **76.1** to **76.4** have the same first distance **D1** and the same second distance **D2**.

[0087] In accordance with a second embodiment, the first sections **T1** of the first and second branches **80**, **82** of the second U-shape preform are positioned between the second sections **T2** of the first and second branches **80**, **82** of the first U-shape preform and connected to those second sections **T2**.

[0088] Of course, the invention is not limited to these embodiments for connecting the U-shape preforms **76.1** to **76.4** to one another.

[0089] In an embodiment that can be seen in FIGS. **15** to **17**, for at least one U-shape preform **76.1** to **76.4** the central part **78** comprises a first part **78.1** corresponding to the body **68** of the blade **64** and a second part **78.2** corresponding to the flange **70**, substantially perpendicular to the first part **78.1** and extending only at the level of the central part **78** of the U-shape preform.

[0090] In accordance with another embodiment that can be seen in FIG. **11**, for at least one U-shape preform **76.1** to **76.4** the second part **78.2** extends over all of the central part **78** and at least partially along at least one of the first and second branches **80**, **82**. In accordance with one arrangement, the flange **70** extends at the level of the central part **78** and over each of the first and second branches **80**, **82**, notably at the level of the first sections **T1** of the first and second branches **80**, **82**. This embodiment enables better resistance to the forces produced on the flange **70** by a flow of air diverted by the grid of blades **60**, those forces being transmitted to the body **68** and the first and second branches **80**, **82**.

[0091] At least one U-shape preform **76.1** to **76.4** has an interior face **F76** including the interior surface **F1** of the body **68** of the central part **78** oriented toward the first and second branches **80**, **82**, the interior surface **F2** of the flange **70** oriented toward the first and second branches **80**, **82**, and the interior surfaces **F3**, **F4** of the first and second branches **80**, **82** oriented toward one another. In accordance with one particular feature of the invention, the interior face **F76** of each U-shape preform **76.1** to **76.4** does not have an undercut shape in an extraction direction **DT** substantially parallel to the longitudinal direction.

[0092] In accordance with one embodiment, the step of producing the U-shape preforms **76.1** to **76.4** comprises for at least one U-shape preform **76.1** to **76.4** and preferably for all of them at least one phase of compression of said U-shape preform **76.1** to **76.4** and at least one curing or polymerization phase during which said U-shape preform **76.1** to **76.4** is compressed and at least partially cured or polymerized.

[0093] In accordance with one variant, the two phase of compression and curing or polymerization

are simultaneous. In accordance with another variant, the curing or polymerization phase is completed before the end of the compression phase. In accordance with this variant, the U-shape preform **76.1** to **76.4** is compressed before the at least partial curing or polymerization is completed.

[0094] By way of example, during the compression and curing or polymerization phases each U-shape preform **76.1** to **76.4** is compressed at a pressure between 2 and 7 bar and heated to a temperature of the order of 180° C. Of course, the invention is not limited to these pressures and this temperature. The latter can vary as a function of the materials of a U-shape preform **76.1** to **76.4**.

[0095] At the end of the production step, each U-shape preform **76.1** to **76.4** is sufficiently cured or polymerized to retain a stable geometry and stable dimensions.

[0096] These compression and at least partial curing or polymerization phases make it possible to guarantee the structural integrity of the grid of blades **60**.

[0097] Before the compression phase, the step of production of the U-shape preforms **76.1** to **76.4** comprises for at least one U-shape preform **76.1** to **76.4** a draping phase during which layers of fibers **88** are positioned on a first mold **86.1** shaped like the interior face **F76** of the U-shape preform **76.1** to **76.4**.

[0098] Tooling **86** for carrying out the production step comprises for each U-shape preform **76.1** to **76.4** to be produced at least one first mold **86.1** on which the U-shape preform **76.1** to **76.4** is positioned at least during the compression and partial curing or polymerization phases.

[0099] Providing for each U-shape preform **76.1** to **76.4** an interior face **F76** with no undercut makes it possible to use a simple first mold **86.1** and to be able to extract the U-shape preform **76.1** to **76.4** from the mold without difficulty.

[0100] In accordance with a first operating mode, during the production step the U-shape preform **76.1** to **76.4** positioned on the first mold **86.1** is compressed using at least one compression bladder. In this case, the tooling **86** comprises in addition to the first mold **86.1** at least one compression bladder.

[0101] In accordance with a second operating mode that can be seen in FIGS. **12** and **1**, during the production step the U-shape preform **76.1** to **76.2** is compressed between the first mold **86.1** and at least one matching mold **86.2**. In this case the tooling **86** comprises in addition to the first mold **86.1** at least one matching mold **86.2**.

[0102] In accordance with a third operating mode, during the production step the U-shape preform **76.1** to **76.2** is compressed between on the one hand the first mold **86.1** and on the other hand at least one matching mold and at least one compression bladder. In this case the tooling **86** comprises in addition to the first mold **86.1** at least one matching mold **86.2** and at least one compression bladder.

[0103] In accordance with one operating mode, the step of production of a U-shape preform **76.1** to **76.4** comprises a step of draping layers of fibers **88** on a placement mold having a placement surface **F90** shaped like the interior face **F76** of the U-shape preform **76.1** to **76.4**. In accordance with one configuration, the placement mold on which the layers of fibers **88** are deposited corresponds to the first mold **86.1**. Of course, the invention is not limited to this embodiment. The placement mold and the first mold **86.1** could be two distinct molds.

[0104] For at least one U-shape preform, during draping, at least one layer of fibers **88** is positioned so as to straddle the first and second parts **78.1**, **78.2** of the central part **78** to obtain a body **68** and a flange **70** in one piece, which contributes to improved transmission of forces between these two parts of the blade **64**.

[0105] For at least one U-shape preform, during the draping phase, the layers of fibers **88** are positioned in such a manner as to feature fibers mostly oriented in the transverse direction at the level of the central part **78** and fibers oriented mostly in the longitudinal direction at the level of each of the first and second branches **80**, **82**.

[0106] To ensure improved transmission of forces between the blade **64** and the longitudinal walls **62, 62'**, the central part **78** is connected to each of the first and second branches **80, 82** with a relatively large radius of curvature so as to ensure improved transmission of forces from the central part **78** to the first and second branches **80, 82**.

[0107] In accordance with one embodiment, for at least one U-shape preform **76.1** to **76.2**, during the draping phase, at least one layer of fibers **88** is positioned in such a manner as to extend over the central part **78** and at least one of the first and second branches **80, 82**. This embodiment makes it possible to improve the transmission of forces from the central part **78** to the first and second branches **80, 82**. At least one layer of fibers **88** is preferably positioned in such a manner as to extend over the first and second parts **78.1, 78.2** of the central part **78** and at least one of the first and second branches **80, 82**.

[0108] In accordance with one arrangement, as depicted in FIG. **12**, at least one layer of fibers **88** of the first branch **80** has a part **88.1** positioned at the level of the central part **78**. The central parts **88.1** of the layers of fibers **88** of the first branch **80** are stacked alternately with the layers of fibers **92** of the central part **78**. At the level of the junctions between the central part **78** and the first branch **80** this arrangement can be duplicated at the level of the junction zone between the central part **78** and the second branch **82**.

[0109] Of course, the invention is not limited to the embodiments that have been described. Thus the composite material U-shape preforms could be replaced by metal U-shape profiled members obtained by pressing.

[0110] Whatever the embodiment, a method obtaining a grid of blades **60** comprises: [0111] a step of producing U-shape elements **76.1** to **76.4**, one for each blade **64**, each U-shape element **76.1** to **76.4** including a central part **78** corresponding to one of the blades **64**, a first branch **80** corresponding to at least a part of the first longitudinal wall **62** and a second branch **82** corresponding to at least a part of the second longitudinal wall **62'**, the central part **78** and the first and second branches **80, 82** forming only one and the same part, [0112] a step of assembling the U-shape preforms **76.1** to **76.4** consisting in connecting the first branches **80** of the various U-shape preforms **76.1** to **76.4** to one another in such a manner as to form the first longitudinal wall **62** and connecting the second branches **82** of the various U-shape preforms **76.1** to **76.4** to one another in such a manner as to form the second longitudinal wall **62'**.

[0113] When the grid of blades is made of a composite material, the U-shape elements correspond to U-shape preforms. When the grid of blades is made of metal, the U-shape elements correspond to U-shape profiled members.

[0114] Producing a grid of blades by assembling U-shape elements makes it possible to obtain blades with large depths while preserving simple tooling because of the geometry with no undercut of the U-shape elements.

[0115] While at least one exemplary embodiment of the present invention(s) is disclosed herein, it should be understood that modifications, substitutions and alternatives may be apparent to one of ordinary skill in the art and can be made without departing from the scope of this disclosure. This disclosure is intended to cover any adaptations or variations of the exemplary embodiment(s). In addition, in this disclosure, the terms “comprise” or “comprising” do not exclude other elements or steps, the terms “a” or “one” do not exclude a plural number, and the term “or” means either or both. Furthermore, characteristics or steps which have been described may also be used in combination with other characteristics or steps and in any order unless the disclosure or context suggests otherwise. This disclosure hereby incorporates by reference the complete disclosure of any patent or application from which it claims benefit or priority.

Claims

- 1.** A method of obtaining a grid of blades of an aircraft thrust reversing device, said grid of blades including at least first and second longitudinal walls oriented in a longitudinal direction and blades positioned between said first and second longitudinal walls, connected to the latter and oriented in a transverse direction, the method comprising the steps of: producing U-shape elements, one for each blade, each U-shape element including a central part corresponding to one of the blades, a first branch corresponding to at least a part of the first longitudinal wall and a second branch corresponding to at least a part of the second longitudinal wall, the central part and the first and second branches forming only one and the same part, and, assembling the U-shape preforms by connecting the first branches of the various U-shape preforms to one another in such a manner as to form the first longitudinal wall and connecting the second branches of the various U-shape preforms to one another in such a manner as to form the second longitudinal wall, wherein at least one U-shape preform has an interior face with no undercut, and wherein the step of producing U-shape preforms comprises, prior to a compression phase and for at least one U-shape preform, a draping phase during which layers of fibers are positioned on a first mold shaped like the interior face of the U-shape preform.
- 2.** The method as claimed in claim 1, wherein each U-shape element is a U-shape preform made of a composite material, wherein the step of producing includes for at least one U-shape preform at least one compression phase and at least one curing or polymerization phase during which said U-shape preform is compressed and at least partially cured or polymerized.
- 3.** The method as claimed in claim 2, wherein each U-shape preform is partially cured or polymerized during the step of producing which comprises a phase of partial curing or polymerization, a phase of final curing or polymerization being carried out subsequently and no later than during the assembling step.
- 4.** The method as claimed in claim 2, wherein each U-shape preform is totally cured or polymerized during the producing step.
- 5.** The method as claimed in claim 1, wherein during the producing step the U-shape preform is compressed between the first mold and at least one matching mold, or at least one compression bladder, or both.
- 6.** The method as claimed in claim 1, wherein during the draping phase, the layers of fibers are positioned in such a manner as to have fibers mostly oriented in the transverse direction at a level of the central part and fibers mostly oriented in the longitudinal direction at a level of each of the first and second branches.
- 7.** The method as claimed in claim 1, wherein, during the draping phase, at least one layer of fibers is positioned in such a manner as to extend over the central part and at least over one of the first and second branches.
- 8.** The method as claimed in claim 1, wherein for each U-shape element the first and second branches are spaced by a distance apart that is substantially constant over a length of the first and second branches, and wherein the distances apart separating the first and second branches of the various U-shape elements decrease from one end to the other of the grid of blades so as to be able to form the first and second longitudinal walls by respectively stacking the first and second branches of the various U-shape elements.
- 9.** The method as claimed in claim 8, wherein the first and second branches of the various U-shape elements have free ends situated at a level of one end of the grid of blades.
- 10.** The method as claimed in claim 1, wherein for at least one of the first and second U-shape elements each of the first and second branches comprises a first section that extends from the central part, a second section that extends from the first section as far as a free end of the first or second branch, and an inclined section connecting the first and second sections, wherein the first sections of the first and second branches of the first and second U-shape elements are spaced by a first distance, the second sections of the first and second branches being spaced by a second

distance greater than the first distance, and wherein the first sections of the first and second branches of the second U-shape element are positioned between the second sections of the first and second branches of the first U-shape element and connected to said second sections.

11. The method as claimed in claim 1, wherein for at least one U-shape element the central part comprises a first part forming a body of a blade and a second part substantially perpendicular to the first part forming a flange of the blade.

12. The method as claimed in claim 11, wherein the second part extends over all the central part and at least partially along at least one of the first and second branches.
